Chassis Dynamometer



Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Chassis dynamometer

What it is used for

A chassis dynamometer measures a vehicle's power and performance by running its driven wheels on rollers or using a hub-connected test system. Correct vehicle setup, secure restraint, ventilation, and controlled test operation help obtain reliable results while reducing vehicle damage and operator hazards.

Main parts

- Rollers or hub adapters transfer vehicle power to the measurement system.
- Drive or load unit applies the required test load where fitted.
- Speed sensors measure wheel or roller speed.
- Torque sensors measure rotational force.
- Control console sets the test programme and displays results.
- Vehicle restraint straps or fixtures secure the vehicle.
- Cooling fans provide airflow during testing.
- Emergency stop and safety guards help control hazards.

Before you start

1. Only trained automotive testing personnel should operate the dynamometer.
2. Check the vehicle manufacturer's limits and confirm that the vehicle and drivetrain are compatible with the test system.
3. Inspect rollers or hub adapters, restraint fixtures, sensors, guards, cables, and controls for damage.
4. Check tyre condition and pressure where roller testing is used, and inspect the test surface for debris.
5. Secure the vehicle using the specified restraints and verify that they are correctly attached.
6. Provide adequate ventilation and cooling airflow, and ensure the exhaust extraction system is working where required.
7. Confirm emergency stops, interlocks, and test limits are set and functional.

How to use

1. Position the vehicle on the dynamometer using the approved alignment procedure.
2. Secure the vehicle with the specified restraints and verify clearance around wheels and moving parts.
3. Connect required measurement leads or diagnostic equipment according to the test plan.

4. Set the test programme, speed limits, load limits, and other parameters within approved vehicle limits.
5. Start the cooling fans and exhaust extraction before running the engine where required.
6. Start the test at the prescribed low load and follow the dynamometer operating instructions.
7. Increase speed or load only as specified by the test procedure while monitoring live readings.
8. Watch engine temperature, vehicle stability, tyre behaviour, abnormal vibration, and warning indicators throughout the test.
9. Stop the test immediately if restraint movement, tyre damage, overheating, unexpected vibration, or another unsafe condition occurs.
10. Reduce load and bring the system to a complete stop before removing restraints or moving the vehicle.
11. Save the test results and inspect the vehicle and equipment for signs of damage.

⚠ Safety

- Never operate a chassis dynamometer without training, correct restraints, and required guards.
- Keep all people clear of rotating wheels, rollers, belts, and vehicle travel paths.
- Do not stand directly behind or in front of a running vehicle during testing.
- Ensure adequate exhaust extraction and ventilation to prevent harmful exhaust-gas accumulation.
- Do not exceed vehicle, tyre, roller, or dynamometer limits.
- Never attach or remove restraints while rollers or wheels are moving; follow the emergency shutdown procedure if the vehicle shifts.

Care & storage

- Remove dirt and debris from the test area after the system is fully stopped.
- Inspect rollers, hub adapters, restraint straps, sensors, guards, and cooling fans regularly.
- Keep measurement software and sensors maintained according to the manufacturer's instructions.
- Arrange authorised calibration and service when measurements are inconsistent or faults appear.

Common mistakes to avoid

- Running a test without verifying vehicle compatibility and restraint security.
- Ignoring tyre condition, vehicle temperature, or dynamometer limits.
- Testing without adequate ventilation or cooling airflow.
- Releasing restraints before the system and vehicle wheels have completely stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.